

RISE Student Research Grant Proposal

Part A: Project Proposal Summary

A.1 Proposed title	
A Multimodal Remote Assessment Framework for Parkinson's Disease Integrating Kinematic Graphs, Masked Foundation Models, and Demographic Profiles	
A.2 Research area	
Health Informatics, Computer Vision, Deep Learning, Spatiotemporal Modeling, Multimodal Machine Learning, Movement Disorders	
A.3 Research summary (max. 150 words)	
Parkinson's Disease (PD) is a rapidly escalating neurodegenerative disorder whose timely assessment is hindered by specialist shortages and geographic disparities. While remote video-based screening offers an accessible alternative, current paradigms face critical bottlenecks: skeleton-based graph convolutions (e.g., PULSAR) degrade across non-Western cohorts due to webcam tracking noise and domain shifts, while Video Foundation Models (VFMs) are vulnerable to background confounders (Clever Hans effect). Furthermore, existing pipelines lack resilience to missing task modalities and omit rich patient clinical-demographic data. This project develops an end-to-end multimodal remote PD screening framework evaluated on a native Bangladeshi cohort. We integrate: (1) an enhanced multi-stream PULSAR architecture with missing-modality imputation across motor (finger-tapping), facial (smiling), and vocal tasks; (2) anatomical ROI-masked VFM fine-tuning to eliminate environmental bias; and (3) a clinical-demographic tabular classifier. By ensembling kinematic, spatiotemporal, and demographic representations, this research delivers an unbiased, clinically interpretable diagnostic solution.	
A.4 Project duration (in months)	10
A.5 Total project cost (in BDT)	1,00,000

Part B: Research Project Details

B.1 Background and statement of the problem (Word limit: Max. 200)
Parkinson's disease (PD) affects over 6.1 million individuals globally, characterized by progressive motor impairments including bradykinesia, tremors, and facial hypomimia [1]. In developing countries like Bangladesh, centralized neurological facilities and severe movement disorder specialist shortages severely impede early diagnosis and longitudinal monitoring [2]. Remote, webcam-based telemedicine screening presents a highly scalable diagnostic pathway [3]. However, current automated screening systems exhibit critical technical gaps. First, state-of-the-art skeleton-based graph convolutional networks such as PULSAR [4] degrade significantly on native South Asian cohorts due to cross-ethnic domain shifts and keypoint extraction noise in unconstrained home webcam environments [Shabab's Thesis]. Second, while Video Foundation Models (VFMs) achieve high empirical accuracy [9], they act as black boxes prone to the 'Clever Hans' effect [5,6,7]—learning spurious background, lighting, and room artifacts rather than true pathological dynamics. Third, speech-based screening is itself susceptible to similar shortcut artifacts [8], and real-world tele-assessments frequently suffer from incomplete submissions (missing motor, facial, or speech tasks), causing existing models to fail. Finally, patient demographic and clinical history biomarkers remain entirely unintegrated.

This research addresses these challenges by developing a multimodal, confounder-free framework combining robust kinematic graph convolutions, masked VFMs, and clinical-demographic profiling under missing-modality constraints.

References:

- [1] Dorsey, E. R., et al. (2018). Global, regional, and national burden of Parkinson's disease. *The Lancet Neurology*, 17(11), 939–953.
- [2] Dorsey, E. R., et al. (2020). Ending Parkinson's Disease: A Prescription for Action. *PublicAffairs*.
- [3] Islam, M. S., et al. (2023). Using AI to measure Parkinson's disease severity at home. *npj Digital Medicine*, 6(1), 156.
- [4] Alam, M. Z. U., Azad, A., Islam, M. S., Hoque, E., & Rahman, M. S. (2026). PULSAR: Graph-Based Positive Unlabeled Learning with Multi-Stream Adaptive Convolutions for Parkinson's Disease Recognition. *ACM Transactions on Computing for Healthcare*, 7(2), 37.
- [5] Lapuschkin, S., et al. (2019). Unmasking Clever Hans predictors and assessing what machines really learn. *Nature Communications*, 10(1), 1096.
- [6] Geirhos, R., et al. (2020). Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11), 665–673.
- [7] DeGrave, A. J., et al. (2021). AI for radiographic COVID-19 detection selects shortcuts over signal. *Nature Machine Intelligence*, 3(7), 610–619.
- [8] Liu, Y.-L., et al. (2024). Clever Hans effect found in automatic detection of Alzheimer's disease through speech. *Interspeech 2024*, 2435–2439.
- [9] Islam, M. S., et al. (2026). Benchmarking video foundation models for remote Parkinson's disease screening. *arXiv:2602.13507*.

Not Referenced:

- [10] Adnan, T., et al. (2023). Unmasking Parkinson's disease with smile: An AI-enabled screening framework. *arXiv:2308.02588*. Mentions High accuracy of 87.9 & AUROC of 89.3. Where can I cite this? (B.3 Maybe?)

B.2 Specific objectives (Word limit: Max. 200)

The primary objectives of this research project are:

1. To adapt and enhance the PULSAR graph convolutional architecture on a novel, native Bangladeshi dataset by **finger-tapping, smiling, and speech tasks**. (Focus Only on Finger Tapping - Sir)
2. To eliminate the Clever Hans effect in Video Foundation Models (VFMs) by developing automated Region-of-Interest (ROI) segmentation and masking pipelines, ensuring spatiotemporal representations isolate true pathological motor dynamics from environmental confounders.
3. To construct a specialized tabular deep learning classifier that leverages patient clinical, lifestyle, and demographic features (e.g., age, neurological family history, smoking history) to extract complementary diagnostic priors.
4. To design an adaptive multimodal fusion and ensemble framework that dynamically aggregates predictions from kinematic graphs, masked VFMs, and demographic networks, maintaining high diagnostic accuracy **even when specific modalities are unavailable**. (Focus Only on Finger Tapping - Sir)

B.3 Expected outcomes (Word limit: Max. 150)

The expected outcomes of this research project are:

1. An improved, domain-adapted PULSAR model capable of resilient kinematic feature extraction with **built-in missing-modality handling** across motor, facial, and vocal tasks.
2. A confounder-resilient VFM fine-tuning framework that demonstrates superior generalizability by eliminating background Clever Hans artifacts through anatomical masking.
3. An interpretable demographic-clinical classification pipeline providing independent baseline risk stratifications from patient history and lifestyle variables.
4. A unified multimodal ensemble architecture demonstrating superior diagnostic efficacy (**achieving robust generalizability with >0.85–0.90 AUROC**), outperforming standalone kinematic graphs (PULSAR) and tabular baselines while provably eliminating background attribution saliency.

5. High-quality peer-reviewed research publications in top-tier conferences or journals (e.g., IEEE/ACM transactions or health informatics venues) advancing remote neurological telemedicine.

B.4 Methodology (Word limit: Max. 500)

Data Acquisition & Multimodal Modalities:

We utilize a novel, resolution-standardized Bangladeshi dataset collected via a web-based telemedicine platform. The dataset comprises multimodal recordings: standardized motor tasks (finger-tapping), facial expression tasks (smiling/hypomimia), speech tasks, and comprehensive clinical-demographic profiles (15+ clinical, lifestyle, and demographic variables including age, neurological family history, smoking, and medications).

Pipeline 1: Enhanced PULSAR with Missing Modality Handling:

We extract 3D hand and facial landmark coordinates using MediaPipe across standard and augmented video streams. An Adaptive Graph Convolutional Network (AGCN) models four kinematic streams: Joint, Bone, Velocity, and Acceleration. To overcome real-world data incompleteness, we implement a modality-dropout and cross-attention imputation layer that enables the network to infer representations even when specific tasks (e.g., facial or voice recordings) are missing. Training incorporates Positive-Unlabeled (PU) loss to mitigate self-reporting label noise in community controls.

Pipeline 2: Debounded & Masked Video Foundation Models (VFMs):

To exploit the rich spatiotemporal feature representation of pre-trained VFMs (e.g., V-JEPA2, TimeSformer, VideoMAE) without falling prey to the Clever Hans effect, we develop an automated anatomical masking pipeline. Dynamic bounding boxes and segmentation masks isolate the anatomical region of interest (hands for tapping; face for smiling), replacing background pixels with neutral values. This forces transformer attention heads to attend strictly to kinematic motion and micro-expressions, eliminating room/lighting biases.

Pipeline 3: Clinical & Demographic Classifier:

We preprocess tabular clinical variables using target-aware encoding, outlier normalization, and missing-value imputation. We deploy a TabNet / Gradient Boosted architecture with attention mechanisms to learn discriminative disease embeddings from clinical and lifestyle risk factors.

Multimodal Fusion & Late Ensembling:

We design an adaptive late-fusion gating mechanism that computes dynamic confidence weights for each pipeline based on modality availability and signal quality. The final diagnostic prediction aggregates kinematic, spatiotemporal VFM, and demographic probabilities.

Challenges & Mitigation:

1. High Computational Demand: Training multi-stream graph convolutions, fine-tuning large video foundation transformers on high-resolution frames, and tabular feature modeling simultaneously require substantial GPU VRAM, rapid SSD throughput, and high system memory. We request dedicated computational resources to facilitate extensive hyperparameter search and cross-validation on-premises without cloud bandwidth bottlenecks.
2. MediaPipe Tracking Noise: Mitigated via temporal smoothing and kinematic interpolation.
3. Label Noise: Mitigated through Positive-Unlabeled (PU) loss formulations.
4. Clever Hans effect: Mitigated through dynamic bounding boxes and segmentation masking.

Part C: Proposed Grant Amount Summary

(Detail budget must be submitted in a separate file.)

SN	Budget description	Total amount

		(BDT)
(i)	Resources	
(ii)	Miscellaneous	
(iii)	Conference Registration and Participation Expenses (maximum BDT 10,000)	10,000
Grand total amount (i+ii+iii)		10,000
(in words)	One Lac Taka Only	

Part D: Student Details

D.1 Applicant-1	
Name	Md. Mobasharul Islam Tonmoy
Department	Department of Computer Science and Engineering
Email address	2105076@ugrad.cse.buet.ac.bd
Contact no.	01976864607
Research experience (Word limit: Max. 100)	1. Co-Built a multi-purpose robotic arm.¹ 2. Co-developed an abstractive news summarizer for Bengali using transformer models² 3. Working on Parkinson's Analysis with Remote Tasks under the supervision of Dr. Mohammad Saifur Rahman, CSE, BUET from July 2026 to date.
D.2 Applicant-2	
Name	
Department	
Email address	
Contact No.	
Research experience (Word limit: Max. 100)	
<i>Add more applicants if necessary.</i>	

¹ <https://github.com/mnh-14/multi-purpose-robotic-arm>

² https://github.com/Diganta81/ML_Project_Abstractive_Bangla_News_Summarizer/tree/main